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Mst Rokaya Akter Rakhi

Software Engineer · AI/LLM Systems & Backend

Final-year Computer Science Engineering student and Stipendium Hungaricum scholar (cumulative average 4.57/5.00). I research how to compare large language models reproducibly, and I build the backend systems that make such experiments reliable. I am seeking a fully funded MSc in Artificial Intelligence / NLP from autumn 2027.

Education

BSc Computer Science Engineering (taught in English) Sep 2023 – expected Feb 2027

University of Pécs, Faculty of Engineering and Information Technology, Hungary

- Cumulative average **4.57 / 5.00** after six semesters (corrected credit index 4.67), with 193 ECTS earned.
- Excellent (5) in Algorithm Design, Software Engineering, Operating Systems, Databases 2, Programming Paradigms, Computer Architectures, Cyber Security, Web Programming 1–2, Engineering Mathematics 1 & 3, and Hackathon (×2).

Higher Secondary Certificate 2019 – 2021

New Govt. Degree College, Rajshahi, Bangladesh

Research

LLM Comparative Framework for Automated Technical Question Generation BSc thesis · 2025 – 2027

Supervisor: Dr. Tamás László Storcz

- Designed a reproducible pipeline that generates technical multiple-choice questions with six LLMs from five providers: GPT-4o, GPT-OSS 120B, Llama 3.3 70B, Mistral Large/Small and Cohere Command A.
- Built deterministic “Three Pillars” scoring (cosine semantic alignment, Flesch–Kincaid complexity, Jaccard distractor plausibility) to avoid AI-judging-AI bias.
- Engineered for reproducibility: a versioned relational store (SQLAlchemy, SQLite WAL), parallel batch runs with jittered backoff, token-cost logging, and local/lab/cloud profiles. Includes an expert-validation study.

EduGen: microservice platform for blind evaluation of LLM-generated questions Oct – Nov 2026

- Four services (Lab, Generator, Evaluator, Review) with RabbitMQ, PostgreSQL, a React client and an nginx gateway on Docker Compose. Blind expert rating and an anonymous student quiz let human judgement be compared with the automatic scores. 146 automated tests, including cross-service contract tests.

Experience

Software Engineering Intern, University of Pécs, Faculty of Engineering and IT Sep 2026 – present

Campus Engage, a gamified university engagement platform · Supervisor: Dr. Tamás László Storcz

- Co-wrote the functional and technical specifications, then delivered 16 backlog tasks across two sprints in a three-person team (65 commits in the first month).
- Built Filament admin tools (invitations, account management, bulk CSV import, session revocation) and the event-application engine (capacity guard, waitlist auto-promotion, anti-self-approval selection).
- Built a QR attendance scanner that credits points through a central ledger, plus queued Neptun enrolment imports, integration tests and demo data. Laravel 13, Vue 3/Inertia, MySQL, Redis, Docker, GitHub Actions.

AI/ML Engineering Intern, DevelopersHub Corporation 2026 · Phases 1 & 2

- Fine-tuned BERT for news topic classification on AG News (89.6% accuracy, 89.57% F1).
- Built a context-aware RAG chatbot (LangChain, FAISS, OpenAI) with query rewriting (80% retrieval accuracy).
- Few-shot vs zero-shot LLM ticket tagging with Pydantic structured outputs: schema adherence rose from 12.5% to 66.7%.
- Built a multimodal PyTorch model (ResNet18 + MLP) and scikit-learn pipelines for churn (≈80.5% acc.), heart-disease risk (ROC-AUC ≈ 0.92) and house prices (MAE ≈ $16.6k).

Selected projects

VolunteerBridge Java 21 · Spring Boot 4

An NGO volunteer and task manager with JWT and role-based access (4 roles), live WebSocket alerts for critical tasks, automatic hour logging and audit trails.

ShelfLife Java 21 · Spring Boot 3.2

A book platform with shelves, reviews and live STOMP notifications. It uses a generic CRUD service layer, Spring Security, GitLab CI and Docker.

Scholarships & honours

- **Stipendium Hungaricum Scholarship**, Government of Hungary / Tempus Public Foundation: full BSc scholarship, 2023 – present.
- **Chinese Government Scholarship** (bilateral programme, Harbin): awarded 2023.
- Stipendium Hungaricum Mentorship Programme: member since Dec 2023.

Skills

Languages	Python, Java, TypeScript/JavaScript, SQL, C/C++
AI & ML	PyTorch, Hugging Face Transformers, scikit-learn, LangChain, LangGraph, RAG, FAISS; OpenAI, NVIDIA NIM, Mistral and Cohere APIs
Backend & data	Spring Boot, FastAPI, REST, WebSockets, RabbitMQ, PostgreSQL, MySQL, SQLite, MongoDB
Tooling	Docker, GitLab CI/CD, nginx, Linux, Git

Leadership & community

- **Core Team Member, MIK Hub**, student innovation hub, Faculty of Engineering and IT, University of Pécs: internship pipeline and makerspace, 2026 – present.
- **Volunteer, Evergreen Society**, Rajshahi: urban tree-planting programme, 2021 – present.
- **Team Leader, Rover Scouts** (House Mujibnagar), New Govt. Degree College, 2019 – 2021.
- Shotokan Karate-do International trainee, since 2022. School English debate, recitation and quiz competitions.

Languages

Bengali (native) · English (C1, degree taught in English) · Hindi (fluent, spoken) · Urdu (spoken) · Hungarian (elementary)